Bean diseases

Bacterial diseases

1. Common bacterial blight/ Fuscous blight -*Xanthomonas axanopodis* pv. *phaseoli*.
2. Halo blight / Greasy spot - *Pseudomonas savastanoi* pv. *phaseolicola*
3. Bacterial wilt - *Curtobacterium flaccumfaciens*

Viral diseases

1. Bean common mosaic - *Bean Common Mosaic Virus* (BCMV)
2. Bean yellow mosaic-*Bean Yellow Mosaic Virus*, *Mungbean yellow virus* or *Phaseolus virus*-2 (ss DNA virus)

Common bacterial blight -*Xanthomonas axanopodis* pv. *phaseoli*

Common bacterial blight -*Xanthomonas axanopodis* pv. *phaseoli*

- Also known as **fuscous blight**
- Occurring more commonly than halo blight with yield losses up to 84% in cvs. contender and arka komal

Symptoms: Stems, branches, leaves, pods and seeds

- The symptoms appear as characteristic water soaked spots on the under side of the leaves that enlarge and coalesce together to form large blighted areas.
- Infected areas often become flaccid and turn brown giving it a sunscald appearance.
- If infection occurs through the vascular tissues, reddish discolouration of veins and midribs is observed.

Common bacterial blight -*Xanthomonas axanopodis* pv. *phaseo*

Symptoms contd…

- In advanced stages, tissues become necrotic and defoliation may also take place.
- Lesions on stems appear as sunken water soaked areas, which gradually enlarge to form reddish streaks extending longitudinally in the stem.
- In early stages of pod formation characteristic stem girdling or joint rot frequently develop
- Infection on pods may occur at any place as small water soaked spots, which are surrounded by distinct zones of narrow reddish brown or brick red band of tissue.
- On seeds, discoloration appears especially near the hilum region.

Common bacterial blight -*Xanthomonas axanopodis* pv. *phaseo*

Pathogen:

- The disease is caused by *Xanthomonas axanopodis* pv. *phaseoli*.
- The bacterium produces diffusible brown pigment in the culture
- The bacterium is a motile, non-capsulated, non-spore forming, gram -ve, rod shaped, measuring 1.9 x 0.8 µm in size with a single polar flagellum.
- The pathogen forms characteristic yellow coloured colonies due to a water insoluble eacotenoid and a mucoid growth on nutrient glucose agar medium.

Common bacterial blight -*Xanthomonas axanopodis* pv. *phaseo*

Disease cycle and epidemiology:

- The bacterium perpetuate in infected seed, crop debris and weed hosts of which contaminated seed has been observed to be the most efficient mode of survival and dispersal over long distances.
- In infected seed on germination, the bacterium infects cotyledons and can then enter the vascular system of the plant.
- Alternatively the bacterium can be splashed to the plants from the plant debris.

Common bacterial blight -*Xanthomonas axanopodis* pv. *phaseo*

Disease cycle and epidemiology contd…

- Once inside the plant, the bacteria invade and cause breakdown of the middle lamella.
- The bacterium is a warm temperature pathogen and causes greatest damage to the plants at 28-32° C.
- High humidity and high rainfall increases the disease spread.
- Secondary spread occurs by wind blown rain (rain splashes), contact between leaves, irrigation water and insects like white flies and leaf miners.

Common bacterial blight -*Xanthomonas axanopodis* pv. *phaseo*

Management:

- Collect and burn the infected plant debris.
- Mixed intercropping with maize
- Follow at least three years crop rotation.
- Avoidance of sprinkler irrigation
- Use healthy seed, i.e the seed produced in arid zones with low rainfall but it's use can not guarantee the clean crop
- Delay in sowing time
- Host resistance interspecific hybrids (5 hybrids)
  P. vulgaris x *P. coccineus*

Common bacterial blight -*Xanthomonas axanopodis* pv. *phaseo*

Management contd…

- Hot water treatment of seed at 50°C/ 10 min followed by dipping in streptocycline (100 ppm) solution.
- With the initiation of the disease spray the crop with copper oxychloride + zineb or copper oxychloride + mancozeb and repeat at 10 days interval.

Halo blight /greasy spot - *Pseudomonas savastanoi* pv. *phaseolicola*

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- Halo blight has been a serious of bean. This is often referred to as greasy spot disease due to the appearance of greasy water soaked lesions on affected organs
- More prevalent in areas where temp. is low hence it is less imp. as compared to common blight

Symptoms

Mostly occurs on foliage.

On leaves

- As small water soaked spots resembling pin pricks under the surface of the leaves.
- The spots turn brown gradually becoming yellowish green.
- The zone of greenish yellow tissues around the spot resembles a halo (2-3 cms) , hence name halo blight.

Halo blight /greasy spot - *Pseudomonas savastanoi* pv. *phaseolicola*

Symptoms contd…

On pods

- Small water soaked pin-pricks on pod surface, gradually enlarging to form dark sunken spots of various sizes. White bacterial ooze appears on the spot.
- Pod lesions differs from leaf lesions by not having halo around the leaf lesions.
- Scattered water soaked spots on primary leaves indicates seeds were contaminated with halo blight bacteria.

Halo blight /greasy spot - *Pseudomonas savastanoi* pv. *phaseolicola*

Symptoms contd…

- Systemic plant chlorosis with leaf yellowing and malformation may also develop without external infection
- At 20 ^{o}C the systemic chlorosis more pronounced and unifrom During infection the pathogen produces phaseolo toxin which in turn produces systemic chlorosis and typical halo in the plant
- Infections are initiated on the lateral veins which progress to the main vein often accompanied by a marked water soaking and reddish discolouration of the interveinal region

Halo blight /greasy spot - *Pseudomonas savastanoi* pv. *phaseolicola*

Symptoms contd…

- On the young stem, sunken, water soaked spots gradually enlarge and appear as reddish streaks, extending longitudinally along the stem
- In advanced stages of infection, the stem epidermis split open and often accompanied by light cream to silver coloured bacterial exudation
- On affected pods brown to red zonate lesions surround water soaked spots
- The bacterium soon invades the tissues and progress to the developing seed often leading to shrivelling, discolouration and even rotting of seed

Distinct symptoms for diagnosis of two blight diseases

- In Halo blight-Yellow halo is broad compared to narrow halo in common blight.
- In halo blight-cream or silver coloured ooze, in narrow-yellowish bacterial ooze.

Halo blight /greasy spot - *Pseudomonas savastanoi* pv. *phaseolicola*

Disease cycle

- Seed borne pathogen, Pathogen also survives in plant debris and volunteer plants
- Under high moisture and low pH , bacteria survive up to 2-3 months
- Bacterium apparently can not survive in colder regions in soil or plant debris

Halo blight /greasy spot - *Pseudomonas savastanoi* pv. *phaseolicola*

Management:

- Collect and destroy the infected plant debris after harvesting immediately and collateral hosts
- Follow long crop rotation for 2-3 years.
- Use of healthy seed
- Avoid irrigation with sprinkler system
- Resistant variety-blue lake, California.
- Seed treatment with hot water at 70°C/ 2hrs for moist seed or 50°C /3hrs for dry seed reduced 72-74% seed infection

Halo blight /greasy spot - *Pseudomonas savastanoi* pv. *phaseolicola*

Management contd…

- Seed treatment with kasugamycin or streptomycin.

- Spraying copper oxychloride (0.2- 0.25%) from seedling to flowering reduced pod infection or application of COC 3 to 4 times after seedling emergence

Bacterial wilt - *Curtobacterium flaccumfaciens*

Bacterial wilt - *Curtobacterium flaccumfaciens*

Symptoms

- Young plants of 5-7.5cm tall are attacked and killed.
- Plants become stunted and unthrifty.
- Base of the petioles becomes flaccid and droop during day time but regain turgidity during night.
- Stems may crack longitudinally causing sudden death in some cases.

Bacterial wilt - *Curtobacterium flaccumfaciens*

Symptoms contd…

- The symptoms are more conspicuous on ripe pods which are olive green in colour.
- The bacteria survives in seed for 5-24 years

Bacterial wilt - *Curtobacterium flaccumfaciens*

Management

Refer to earlier notes of bacteria.

Bean common mosaic/ green mosaic - *Bean Common Mosaic Virus* (BCMV)

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Distorted leaf

Infection of leaf mostly along the veins

Bean common mosaic/ green mosaic - *Bean Common Mosaic Virus* (BCMV)

Old leaves showing yellow and green patches - mosaic

Mosaic symptoms typical of infection

Bean common mosaic/ green mosaic - *Bean Common Mosaic Virus* (BCMV)

Symptoms :

- Well defined dark green areas of foliar tissues against light green background of affected leaf lamina
- In early stages of infection, affected leaves crinckled, chlorotic, stiff and accompanied by downward curling
- In advanced stages, reduction in leaf size with sinuous or filiform leaflets
- Plants stunted, dwarfed and bushy
- Infected seed gives characteristic symptoms on seedlings, mottling, blistering, downward cupping, chlorosis on the fruit and succeeding compound leaflets

Bean common mosaic/ green mosaic - *Bean Common Mosaic Virus* (BCMV)

Symptoms contd…

- Such plants usually fail to set pods or develop undersized pods
- Pods on severely infected plants are usually undersized and contain fewer ovules than those produced on the normal plants
- Such pods are occasionally covered with small, dark green spots and often mature very late
- The seed may be aborted, smaller and malformed
- The infection causes delayed seed maturity

Bean common mosaic/ green mosaic - *Bean Common Mosaic Virus* (BCMV)

Symptoms contd…

- Roots of affected plants turn dark and during periods of severe infection the infection the tap root system almost turns black due to vascular necrosis giving rise to typical black root symptoms

- Plants affected by black root, initially exhibit slight wilting of the leaflets near the top of the plants which subsequently loose their bright green colour and appear greyish green

- Permanent general wilting of all the leaves follows often resulting in death of the plants

Bean common mosaic/ green mosaic - *Bean Common Mosaic Virus* (BCMV)

Pathogen:

- The disease is caused by *Bean Common Mosaic Virus* (BCMV) which belongs to poty virus group.
- The virus particles are filamentous flexuous rods measuring 700-800 nm in length and 12-15 nm in diameter.
- The genome consists of unipartite, single stranded DNA.

Bean common mosaic/ green mosaic - *Bean Common Mosaic Virus* (BCMV)

Disease cycle and epidemiology:

- The virus is seed borne in a range of legumes . The virus was found to survive in seed coat, cotyledons and embryo as well as in pollen
- Seed-borne inoculum contributes to primary spread, and secondary spread occurs through **several aphid species** in nonpersistent manner
- The virus is also sap and graft transmissible.
- The virus can be transmitted in the non-persistent manner by several aphid species which do not normally colonize beans but transmit the virus as winged migrants, especially *Acyrthiosiphon pisum*, *Aphis fabae* and *Myzus persicae*.

Bean common mosaic/ green mosaic - *Bean Common Mosaic Virus* (BCMV)

Management:

- Use healthy and certified seed grown under screen / green house conditions helps in checking the disease.
- Manipulation of plant population density, closer spacing interfere with vector movement
- Early planting when the incidence of aphid vectors is low gives better yields.
- Rogueing the infected plants
- Planting non-host barriers like jowar/maize
- Soil application of insecticides followed by 2-3 foliar sprays against insect vectors

Bean yellow mosaic - *Bean Yellow Mosaic Virus* (BYMV)

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Bean yellow mosaic - *Bean Yellow Mosaic Virus* (BYMV)

Symptoms:

- Several strains of viruses are present, symptoms vary with variety and strain
- Mosaic symptoms consists of contrasting dark and yellowish green areas often accompanied by bright yellow spots which intensifies with the age of the plant
- Some isolates cause diffuse chlorotic mottle and limited plant stunting, while others prominently cause mosaic, rugosity, malformation and severe stunting
- Some varieties may show necrotic spots, veinal and apical necrosis, wilting and premature death
- Late infection generally causes prominent foliar symptoms

Bean yellow mosaic - *Bean Yellow Mosaic Virus* (BYMV)

Symptoms contd…

- If the infection coincides with pod formation, the pods eventually will exhibit a light green mottle and slight malformation.
- BYMV infected plants are usually stunted and bushy due to reduction in internode length and proliferation of lateral branches.
- Plant maturity is delayed, quality and quantity of the seed is affected
- Certain strains of BYMV induce systemic necrosis that appear as purplish discolouration at the base of the lower leaves, accompanied by veinal, stem and petiole necrosis

Bean yellow mosaic - *Bean Yellow Mosaic Virus* (BYMV)

Symptoms contd..

- The necrosis at the terminal growing point consequently result into death of the plant
- Some strains incite local lesions and other produce malformed pods bearing reddish brown necrotic spots

Bean yellow mosaic - *Bean Yellow Mosaic Virus* (BYMV)

Disease cycle

- Bean yellow mosaic virus (BYMV) belongs to **potyvirus group family Potyviridae**
- Monopartite , positive-sense, single-stranded RNA a filamentous particle that measures about 750 nm in length
- The virus had wide host range can infect 233 spp in 71 genera of 21 families
- Transmission through sap and by aphids (*Aphis fabae*, *A. craccivora*, *Myzus persicae*, *Macrosiphum euphorbiae*)
- Not seed borne in french bean, but seed borne in other leguminous hosts

Bean yellow mosaic - *Bean Yellow Mosaic Virus* (BYMV)

Management

- Destruction of collateral hosts
- Growing barrier crops like maize
- Crop rotation
- Rest. Varieties. GNUI No. 31,59,123 and 1140
- Spraying with insecticides for vector management

Bean diseases

Fungal diseases

1. Cercospora leaf spot - *Cercospora cruenta*
2. Rust - *Uromyces phaseoli typica*, *U. appendiculatus*
3. Powdery mildew - *Erysiphe polygoni*
4. Anthracnose - *Colletotrichum lindemuthianum*
5. Stem and pod anthracnose *Colletotrichum truncatum*
6. Angular leaf spot - *Phaeoisariopsis griseola*
7. Charcoal rot or dry root rot- *Macrophomina phaseolina*
8. Root rot and web blight- *Rhizoctonia solani*